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Newton polytopes of immanants of some combinatorial matrices.

For an $n \times n$ matrix $A = (a_{ij})$ and a partition $\nu \vdash n$, the *immanant* of A is defined as

$$\mathrm{Imm}_\nu(A) = \sum_{\pi \in S_n} \chi^\nu(\pi) \prod_{i=1}^{n} a_{i,\pi(i)},$$

where χ^ν denotes the irreducible character of S_n indexed by ν. For a skew partition λ/μ of length n, the *Jacobi–Trudi matrix* is $H(\lambda, \mu) = (h_{\lambda_i - \mu_j - i + j})_{i,j=1}^{n}$, where h_r is the complete homogeneous symmetric function of degree r. For a partition λ of rank k, with Frobenius notation $(\alpha_1, \ldots, \alpha_k \mid \beta_1, \ldots, \beta_k)$, the *Giambelli matrix* is $G_\lambda = (s_{(\alpha_i \mid \beta_j)})_{i,j=1}^{k}$, where the entries are Schur functions of hook shape.

The *Newton polytope* of a polynomial is the convex hull of the exponent vectors of its monomials with nonzero coefficients. A polynomial has a *saturated Newton polytope* (SNP) if every integer point in its Newton polytope belongs to its support. *Monical–Tokcan–Yong* [Sel. Math., New Ser. 25, No. 5, Paper No. 66, 37 p. (2019; Zbl 1426.05175)] showed that Schur polynomials have saturated Newton polytopes, which are permutahedra. The paper under review investigates this property for immanants of Jacobi–Trudi and Giambelli matrices.

For a skew partition λ/μ of size d and length n, the author proves that $\mathrm{Imm}_{(n)}\, H(\lambda, \mu)$ is SNP. The same holds for $\nu = (n-1, 1)$ when λ/μ is a border strip. The author also proves that the symmetric functions $E^{\theta}_{\lambda/\mu}$, introduced by *Stanley–Stembridge* [J. Comb. Theory, Ser. A 62, No. 2, 261–279 (1993; Zbl 0772.05097)] in their study of Jacobi–Trudi immanants, are SNP for every partition $\theta \vdash d$ whenever λ/μ is a border strip. For Giambelli matrices, the author proves that every immanant is SNP, with Newton polytope the permutahedron of a partition determined by λ and ν.

The proofs use lattice-path interpretations of immanants, together with character sums and the relationship between dominance order and permutahedra. The author conjectures that $\mathrm{Imm}_\nu\, H(\lambda, \mu)$ is SNP for every skew partition λ/μ of length n and every $\nu \vdash n$. Further conjectures state that the normalizations of the Giambelli immanants, and of $E^{\theta}_{\lambda/\mu}$ when λ/μ is a border strip, are Lorentzian polynomials. The author shows by an example that the normalization of an immanant of a Jacobi–Trudi matrix need not be Lorentzian.

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MSC:

05B20
05A05
05E05
52B05
05A15

Keywords:

Newton polytope; immanant; lattice path; Jacobi–Trudi matrix; Giambelli matrix; Schur positivity; permutahedron

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